Chorify: A Phygital Approach to Fair and Engaging Chore Distribution

A User Study on Fairness, Engagement, and Social Interaction in a Phygital Chore Distribution System

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Abstract

Household chores are a persistent source of conflict, inequality, and invisible cognitive burden. Yet most existing tools designed to help manage them either ignore these dynamics entirely or quietly reinforce them. This paper presents Chorify, a phygital application that combines physical task cards with a digital mobile interface to make chore distribution fairer and more engaging through gamification. Rather than using game mechanics as simple rewards, Chorify embeds them into the negotiation of household responsibilities through randomised task assignment, peer challenges, skill and luck-based games, and social recognition features. A pilot user study with eight participants found that the system improved perceived fairness and transformed chore management into an unexpectedly social and enjoyable experience, with participants describing it as a bonding time rather than a burden. Mental load reduction was encouraging but inconclusive, pointing to the need for longer term evaluation. These findings suggest that thoughtfully designed gamification can reframe domestic labour as a shared and negotiable activity, opening new directions for technology that addresses household inequality at its social roots.

Introduction

Managing a household is more complex than it appears. Beyond the visible tasks of cooking, cleaning, and childcare lies a persistent and largely invisible layer of work involving the planning, coordinating, and remembering that ensures everything gets done. Researchers refer to this as mental load, and it represents one of the most underacknowledged dimensions of domestic inequality [1, 2]. In most households, this burden is not shared equally. Women disproportionately carry the cognitive and emotional weight of household organisation, often without recognition from the people they live with [3]. The effects extend beyond the home, affecting wellbeing, productivity, and interpersonal relationships [1, 3]. Cultural norms around gender and domestic responsibility are deeply ingrained, and recognising an imbalance does not automatically lead to addressing it [3].

Existing household organisation applications have yet to meaningfully address these dynamics. Rather than challenging the unequal dynamics of domestic labour, most apps quietly reinforce them, positioning one user as the household manager, offering no tools for surfacing mental load, and reducing chore management to simple task lists [4]. Where gamification has been introduced, it tends toward individual competition rather than shared fairness, addressing engagement without addressing equity [4].

We argue that gamification, when thoughtfully designed around social interaction and fairness rather than individual performance, has the potential to do more. Research on gamified systems shows that peer recognition, reciprocal engagement, and shared social norms are among the strongest drivers of sustained behavioural change [5]. These are precisely the dynamics that matter in a household context.

With this in mind, we present Chorify, a phygital application that combines physical task cards with a digital mobile interface to support fairer chore distribution among household members. Chorify uses gamification not as a reward mechanism but as a means of making the distribution of domestic labour visible and negotiable, through randomised task assignment, peer-initiated swapping, skill and luck based challenges, and social recognition features. This work is guided by the research question: does gamifying chores increase perceived fairness in chore distribution and reduce mental load?

To investigate this question, we conducted a user study with eight participants across three sessions, collecting both quantitative and qualitative data on perceived fairness, mental load, emotional response, perceived agency, and learnability.

Related work

2.1 Social Aspects of Household Labor

The unequal distribution of domestic labour remains a persistent feature of modern households. Despite the rise of dual-earner couples, women continue to perform a disproportionate share of unpaid domestic work, a phenomenon Hochschild and Machung [2] termed the "second shift." This inequality extends beyond physical tasks. Barigozzi et al. [1] highlight the cognitive and emotional dimensions of this burden, termed mental load, which includes the planning, anticipating, and coordinating that keeps a household functioning. Their findings reveal a significant perception gap: women consistently report bearing greater organisational responsibility than their partners acknowledge, with 27% of couples showing systematic under-recognition of women's contribution.

The relational consequences of this imbalance are equally significant. Cerrato and Cifre [3] argue that domestic tension arises not from the volume of chores but from the failure to acknowledge their unequal distribution, while Yago and Martínez [6] found that many women do not perceive this inequality as unjust due to internalised gender norms. Rechberger [4] shows that existing household apps largely reproduce these dynamics, positioning women as default managers and ignoring mental load entirely. Chorify addresses this gap by making both physical tasks and organisational contributions visible and negotiable to all household members.

2.2 Gamification in Household Management

Given these social costs, researchers have explored gamification as a way to make chore management engaging and fair. Deterding et al. [7] define gamification as the use of game design elements in non-game contexts, and Hamari et al. [8] emphasise that its success depends heavily on social context rather than mechanics alone. Gamification is not inherently motivating. Its effectiveness emerges from how mechanics interact with the people using them and the relationships between them.

Existing chore applications illustrate both the promise and the limitations of this approach. Kajibuntan [9] offers couples a transparent system for dividing chores based on fair division theory, but relies on purely digital interaction with no gamification mechanics, reducing chore management to a negotiation tool rather than an engaging shared experience. Bjering et al. [10] similarly found that family chore apps rely on extrinsic motivation and individual use, with limited support for the collaborative dynamics that sustain long term engagement. Poorly designed gamification can also backfire. Diefenbach and Müssig [11] identified counterproductive effects in Habitica, and Kappen et al. [12] found gamified task managers did not outperform conventional ones.

When designed around social interaction however, gamification can serve as a conflict resolution tool. Robson et al. [13] caution that unintended behaviours must be anticipated through ongoing design adjustment, while Rumsamrong and Chiou [14] demonstrate that embedding disputes within game mechanics reduces resentment and produces fairer outcomes. Chorify builds on this by embedding

gamification into the chore assignment process through peer-initiated challenges, luck and skill-based games, and mutual recognition features, positioning play not as a reward mechanism but as a tool for negotiating fairness.

2.3 User Behaviour and Habit Formation

Sustaining engagement beyond initial novelty is one of the central challenges in behaviour change systems. Household chores are particularly prone to procrastination as they are repetitive, low in immediate reward, and easy to postpone [15]. Sailer et al. [16] demonstrated that different game elements satisfy different psychological needs as defined by Self-Determination Theory: feedback-based elements like badges increase feelings of competence, while social elements such as teammates foster relatedness. Neither type alone is sufficient, suggesting that effective gamification must combine complementary elements rather than rely on isolated mechanics.

The broader literature reinforces this. Pereira et al. [15] found in a systematic review of 24 studies that the field is dominated by extrinsic motivators, despite intrinsic motivators like narrative and social interaction being more effective for long term behaviour change. Poorly designed systems can even reward procrastination, as seen in Habitica [11]. Kirchner-Krath et al. [17] add that gamification's most valuable contribution is specifically strengthening self-control, suggesting it reframes task engagement as rewarding rather than burdensome. Siriaraya et al. [18] further show that for tedious tasks specifically, player progression mechanics help reduce emotional fatigue and sustain engagement over time.

Ami et al. [19] demonstrate one practical response, combining structured task management, gamification, and social leaderboards to sustain productive engagement. This points toward the value of layering multiple motivational strategies rather than relying on any single element. Chorify draws on these findings by combining randomised task assignment to reduce decision fatigue, skill and luck-based challenges to introduce intrinsic engagement, and peer recognition to activate social relatedness, targeting the psychological barriers to domestic task engagement at multiple levels**.**

3. System Design

Chorify is a phygital application designed to support fair and engaging chore distribution among people who live together, including couples, families, and roommates. Rather than reducing household management to a purely digital experience, Chorify uses physical task cards as the primary interaction point, paired with a mobile application that handles task tracking, game mechanics, and social recognition. Physical objects create a more tangible and shared experience, making chore distribution a visible and collective moment rather than a silent notification on a phone screen. In the full system, cards are embedded with NFC technology, allowing users to interact with the app by tapping their phone to the card. For the purposes of this study, QR codes were used as a functional substitute due to implementation constraints.

At the start of a chore cycle, each household member picks a card from a face-down pile, randomly assigning them a task. This randomisation was deliberate, drawing on Barigozzi et al. [1] who found that the cognitive burden of deciding who does what is itself a significant source of mental load. If a user wishes to avoid their assigned task, they can initiate a swap or a challenge with another household member. A swap is a straightforward exchange of cards. A challenge introduces a game element where the app randomly determines whether the game is skill or luck based. Skill based games include reflex tests, trivia, or simple math challenges, while luck based games are chance driven. The winner skips their chore while the loser completes both tasks.

Once a task is completed, the user can receive a peer review and be awarded a badge by other household members. Hamari and Koivisto [5] found that social recognition and reciprocal benefit are among the strongest drivers of sustained engagement, which motivated this design choice. The app tracks all activity including completed tasks, challenges, exchanges, and badges, giving all household members a shared view of contributions over time.

4. Methodology

This section describes the user study conducted to evaluate Chorify, assessing whether a phygital gamification approach can increase perceived fairness in chore distribution and reduce mental load among cohabiting adults.

4.1 Research Question and Hypotheses

This study addresses the following research question: does using Chorify increase perceived fairness in chore distribution and reduce mental load among people who live together?

We propose three hypotheses. H1: participants will report higher perceived fairness after using Chorify. H2: participants will report lower perceived mental load after using Chorify. H3: participants will find Chorify learnable after a single guided session.

4.2 Participants

Eight participants were recruited through convenience sampling. This sample size is consistent with formative HCI user studies, and the study is framed as a pilot evaluation rather than a definitive assessment. Participants were required to live with at least one other person, be aged 18 or above, and communicate in English or French. People living alone, minors without parental approval, and those with visual impairments were excluded.

4.3 Procedure

The study consisted of three sessions of 10 to 15 minutes each, conducted both in a classroom setting and in participants' own living spaces. Location variation is acknowledged as a limitation. All participants signed a consent form and received a scenario before the first session began.

Session 1 was guided by the research team, introducing participants to Chorify and its features. Participants were asked to think aloud throughout, verbalising their thoughts and difficulties while researchers took observational notes. Session 2 took place without researchers present, allowing observation of natural unguided behaviour and reducing observer bias. Session 3 was held two days later to assess retention and lasting impressions.

The interaction flow was consistent across sessions. Each participant picked a card from a face-down pile, randomly assigning them a chore. Scanning the card's QR code registered the task in the app. To challenge a task, a participant could scan another participant's card, initiating a contest for that chore. The app randomly determined whether the game would be skill based or luck based. The winner skipped their chore while the loser completed both tasks. Participants could also swap cards by exchanging them physically and rescanning. Completed tasks could be acknowledged through peer reviews and badges within the app. At the end of the final session, participants completed a written questionnaire.

4.4 Data Collection and Analysis

Data was collected through researcher observational notes, a post-study questionnaire including Likert scale items on perceived fairness, agency, and learnability, a dedicated Likert scale assessing whether the system reduced participants' cognitive burden related to chores, and app logs tracking tasks, challenges, exchanges, and badges. In cases of missing data, partial responses were retained where usable.

Quantitative data will be analysed descriptively given the small sample size. Qualitative data from observations and open-ended responses will be analysed using inductive content analysis, identifying themes related to fairness, engagement, learnability, and emotional response.

4.5 Ethical Considerations

Participants were informed of the study's purpose and could withdraw at any point. Data was anonymised immediately using unique participant IDs. Researchers had no access to personal identifying information. All data will be stored securely and deleted after one year.

5. Results

This section presents findings from the post-study questionnaire completed by all eight participants, combining quantitative Likert scale responses with qualitative open-ended feedback.

Prior to the study, participants reported significant imbalance in their current household arrangements. Only 12.5% described their chore distribution as equally shared, while 62.5% reported that one person does most of the tasks and 37.5% described their situation as unorganised. Satisfaction with existing arrangements was mixed, and half of participants expressed a desire to change their current system for something more equitable. This confirmed the relevance of the problem Chorify aims to address.

In terms of perceived fairness, results were largely positive. 87.5% of participants agreed or strongly agreed that the chore distribution felt fair during the study. The randomised card assignment was frequently cited as a key factor, with participants noting that having the same number of cards meant everyone had an equal workload. Challenge outcomes and the exchange feature were also well received, with 87.5% agreeing that both felt fair and reduced conflict. Responses regarding mental load were moderately positive, with 62.5% agreeing that the system reduced their cognitive burden related to chores and 37.5% remaining neutral. No participant disagreed. Emotional response data supports this trend, with only 12.5% reporting stress, no participants reporting anger, and 75% feeling more satisfied, less stressed, and more motivated compared to their usual chore organisation system.

Engagement and enjoyment were notably high. 75% agreed that the system made chores more enjoyable and 87.5% agreed that the challenges made the experience more fun. The most commonly reported emotions were enjoyment and satisfaction, both at 62.5%. Participants described the experience as feeling like a "games Friday night" and a "bonding time", and several highlighted the games and challenges as their favourite feature. The system was also perceived as learnable, with 75% agreeing it was easy to use and understand, and 50% strongly agreeing they felt in control of their assigned tasks.

Qualitative feedback was overwhelmingly positive overall, though one participant offered a critical observation, noting that long term effectiveness would depend on sustained regularity and that additional incentives may be needed to prevent users from disengaging over time. This aligns with concerns raised in the literature regarding novelty effects in gamified systems.

6. Discussion & Limitations

The results offer partial support for both hypotheses. Perceived fairness improved clearly, while mental load reduction was less conclusive.

The strongest finding was around fairness. By removing the decision of who does what, Chorify eliminated one of the most conflict generating moments in household management. This is consistent with Barigozzi et al. [1], who found that the cognitive burden of organising and assigning tasks is itself a significant source of mental load. Participants may not have consciously connected the relief of randomisation to a reduction in cognitive burden, which could explain the more neutral responses on the mental load item.

The challenge mechanic produced an unexpected finding. Participants reported enjoyment rather than frustration from losing, suggesting that reframing domestic conflict as structured play genuinely worked. This supports Rumsamrong and Chiou [14], and aligns with Hamari and Koivisto [5] who found that reciprocal social engagement is one of the strongest drivers of sustained motivation. Chores became a shared activity rather than an individual obligation.

One concern raised by a participant deserves attention: long term effectiveness depends on regularity and novelty may fade. Pereira et al. [15] note that up to 80% of gamified systems fail due to poor long term design, and a longitudinal study would be needed to properly evaluate Chorify's lasting impact. It is also worth acknowledging that Chorify can reduce friction around chore distribution but cannot alone address the deeper social norms that sustain domestic inequality, and cultural change remains a necessary complement to any technological intervention [3].

This study has several limitations. The sample of eight participants makes generalisation difficult and the study should be treated as a pilot evaluation. Recruitment through the researchers' own social networks introduces selection bias and limits cultural diversity, as attitudes toward chore distribution vary significantly across cultures. Location variation between sessions, short study duration, and reliance on self-reported data are additional factors that may have influenced results.

7. Conclusion

This paper presented Chorify, a phygital application that combines physical task cards with a digital mobile interface to promote fairer and more engaging chore distribution among cohabiting adults. Motivated by persistent evidence of domestic labour inequality and the failure of existing household applications to address its social and cognitive dimensions, Chorify introduced gamification not as a reward mechanism but as a tool for making household responsibilities visible, negotiable, and enjoyable.

A pilot user study with eight participants provided encouraging preliminary evidence. Perceived fairness improved noticeably, driven largely by the randomised task assignment and the challenge mechanic, which participants experienced as fair and engaging rather than frustrating. Enjoyment and satisfaction were the dominant emotions reported, and several participants described the experience as transforming chores into a shared social activity. Mental load reduction was more modest, suggesting that short term interaction with the system is insufficient to address a burden that is deeply habitual and culturally embedded.

These findings suggest that gamification, when designed around fairness and social interaction rather than competition, can change how people experience chore distribution. Long term effectiveness remains an open question. Future work should focus on testing Chorify over longer periods, with more diverse participant groups, and on developing features that keep users engaged beyond the initial novelty of the system.

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